QPCODE **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) THEORY EXAMINATION MARCH 2026
IV SEMESTER
DSE – MG4DSEPHY202 – EXPLORING THE COSMOS: OBSERVATIONS, CELESTIAL BODIES, AND COSMIC EVOLUTION
(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions Answer any **10** questions.
Each question carries **2** mark

1.	Briefly explain the equatorial coordinate systems used in astronomy.	U/ CO1
2.	Explain the concept of circumpolar stars.	U/ CO1
3.	Define Parsec. Calculate the numerical value of one Parsec.	A/ CO2
4.	Distinguish between flux and luminosity of a star.	U/ CO2
5.	What is meant by the magnification of a telescope? How can we improve it ?	U,An/ CO3
6.	What are the main sequence stars? What is the source of energy that lets the main sequence stars shine?	U/ CO4
7.	The spectrum of a star is observed to peak at 241.5 nm. Determine the effective temperature of the star.	A/ CO4
8.	Write down the steps in the p-p chain reaction.	U/ CO4
9.	What is a black hole? Define the Schwarzschild radius of a black hole.	U/ CO4
10.	A star appears red but has high luminosity. What can you infer about its size and evolutionary stage?	An/ CO4
11.	Do the pulsars pulsate? Justify your answer.	An/ CO4
12.	State and discuss Hubble's law.	U/ CO5
13.	What is meant by redshift? Give its basic formula.	U/ CO5
14.	Mention any two observational evidence supporting the Big Bang model.	U/ CO5

Part B

Short Essay Type Questions
Answer any **6** questions
Each question carries **5** marks

15.	With the help of a labelled diagram, explain the celestial sphere and define the related terms.	U/ CO1
16.	Distinguish between apparent magnitude and absolute magnitude. Calculate the absolute magnitude of a star with apparent magnitude -3.4, which is at a distance of 30 Parsec from Earth.	A/ CO2
17.	Explain the principle, construction, and advantages of reflecting telescopes.	U/ CO3
18.	What are the characteristics of Black Body radiation? Discuss the significance of Planck's radiation law in understanding stellar spectra.	An/ CO4
19.	Describe the life cycle of a star from its birth to its final stage. Explain how stellar mass determines its fate.	U,An/ CO4
20.	Explain Hubble's classification of galaxies based on the morphology.	U/ CO4
21.	Discuss the Standard Big Bang Model of the Universe.	U/ CO5
22.	A Cepheid has a period of 10 days and apparent magnitude 20. Using the Leavitt Law, find its absolute magnitude and distance.	A/ CO5